

MAVLink 2.0 Attack Lab

Step-by-Step Run Guide — All 5 Attacks + Live Detector

Overview

You need **4 terminal windows** open at the same time. Start them in order — each depends on the previous one being ready.

Terminal	What runs	Start when
1	ArduPilot SITL (sim_vehicle.py)	First — always
2	MAVProxy	After SITL prints "EKF2 IMU0 is using GPS"
3	Anomaly Detector (detector.py)	After MAVProxy is connected
4	Attack scripts (one at a time)	After drone reaches 20 m altitude

Terminal 1 — Start SITL

Step 1. Open a terminal and run:

```
# Terminal 1
cd /home/jatin/ardupilot
python3 Tools/autotest/sim_vehicle.py -v ArduCopter --no-rebuild --console
```

Step 2. Wait — SITL takes ~60 seconds to boot. Watch for this line before doing anything else:

```
# Wait for this output
APM: EKF2 IMU0 is using GPS
```

Do NOT open Terminal 2 until you see this line.

Terminal 2 — Start MAVProxy

Step 3. Open a new terminal and run:

```
# Terminal 2
mavproxy.py --master tcp:127.0.0.1:5760 --out udp:127.0.0.1:14550
```

Step 4. Once connected, type these commands one at a time in the MAVProxy prompt (MAV>):

Command	Purpose
param set GPS1_TYPE 14	Enable GPS_INPUT module (JSON over UDP)
module load GPSInput	Load the GPSInput MAVProxy module
signing setup key	Generate MAVLink 2.0 signing key
signing setup sign_outgoing 1	Enable outgoing packet signing

Attack 4 — Replay Attack 1 packet verbatim

```
# python3 attacks/attack4_replay.py  
python3 attacks/attack4_replay.py
```

What to Expect in Terminal 3 (Detector Alerts)

Attack	Script	Expected Alert Tag	What Detector Catches
Attack 1	attack1_signed_cmd.py	ATK1_FUTURE_TS	Sig timestamp >1h in future
Attack 2	attack2_gps_spoof.py	ATK2_GPS_SPOOF	GPS clock >1 day ahead of system
Attack 3	attack3_overflow_dos.py	ATK3_OVERFLOW	GPS time after year 2050
Attack 3B	attack3b_underflow_dos.py	ATK3B_UNDERFLOW	GPS time before Jan 1 2015
Attack 4	attack4_replay.py	ATK4_REPLAY + ATK1_FUTURE_TS	Duplicate hash + future ts

All alerts are also saved to: **detection/results/alerts.log**

Port Architecture

Port	Protocol	Direction	Used by
TCP 5760	MAVLink	SITL → MAVProxy	Internal connection
UDP 14550	MAVLink 2.0	MAVProxy → All	Attacks 1, 4 + Detector
UDP 25100	GPS_INPUT JSON	Attacks → MAVProxy	Attacks 2, 3, 3B + Detector

Quick Troubleshooting

Problem	Fix
MAVProxy: link 1 down	SITL not started yet — start Terminal 1 first
ardupilot: binary not found	Run: <code>mv ~/.local/share/Trash/files/ardupilot ~/ardupilot</code>
Detector: no packets showing	Check MAVProxy is running and drone is armed + airborne
Attack script: Connection refused	MAVProxy not running or GPSInput module not loaded
sim_vehicle.py: command not found	Use full path: <code>python3 /home/jatin/ardupilot/Tools/autotest/sim_vehicle.py</code>